

NAME : GALLA MOHAN SAI

1) Spring Data JPA - Employee Management Example

application.properties

Application Logs

logging.level.org.springframework=info

logging.level.com.cognizant=debug

Hibernate SQL Logs

logging.level.org.hibernate.SQL=trace

logging.level.org.hibernate.type.descriptor.sql=trace

Log pattern

logging.pattern.console=%d{dd-MM-yy} %d{HH:mm:ss.SSS} %-20.20thread %5p %25.25logger{25} %25M %4L %m%n

Database Configuration

spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver

spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn

spring.datasource.username=root

spring.datasource.password=root

Hibernate Configuration

spring.jpa.hibernate.ddl-auto=validate

spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.MySQL8Dialect

MySQL Table

CREATE DATABASE countrydb;

USE countrydb;

CREATE TABLE country (
 code VARCHAR(2) PRIMARY KEY,
 name VARCHAR(50)
);

INSERT INTO country VALUES ('IN', 'India');

INSERT INTO country VALUES ('US', 'United States of America');

Country.java

```
package com.cognizant.ormlearn.model;
```

```
import jakarta.persistence.Column;
```

```
import jakarta.persistence.Entity;
```

```
import jakarta.persistence.Id;
```

```
import jakarta.persistence.Table;
```

```
@Entity
```

```
@Table(name = "country")
```

```
public class Country {
```

```
    @Id
```

```
    @Column(name = "code")
```

```
    private String code;
```

```
    @Column(name = "name")
```

```
    private String name;
```

```
    public String getCode() {
```

```
        return code;
```

```
    }
```

```
    public void setCode(String code) {
```

```
        this.code = code;
```

```
    }
```

```
public String getName() {  
    return name;  
}
```

```
public void setName(String name) {  
    this.name = name;  
}
```

```
@Override  
public String toString() {  
    return "Country [code=" + code + ", name=" + name + "];"  
}  
}
```

CountryRepository.java

```
package com.cognizant.ormlearn.repository;
```

```
import org.springframework.data.jpa.repository.JpaRepository;
```

```
import org.springframework.stereotype.Repository;
```

```
import com.cognizant.ormlearn.model.Country;
```

```
@Repository
```

```
public interface CountryRepository extends JpaRepository<Country, String> {  
}
```

CountryService.java

```
package com.cognizant.ormlearn.service;
```

```
import java.util.List;
```

```
import org.springframework.beans.factory.annotation.Autowired;
```

```

import org.springframework.stereotype.Service;

import org.springframework.transaction.annotation.Transactional;

import com.cognizant.ormlearn.model.Country;

import com.cognizant.ormlearn.repository.CountryRepository;

@Service

public class CountryService {

    @Autowired

    private CountryRepository countryRepository;

    @Transactional

    public List<Country> getAllCountries() {

        return countryRepository.findAll();

    }

}

```

OrmLearnApplication.java

```

package com.cognizant.ormlearn;

import java.util.List;

import org.slf4j.Logger;

import org.slf4j.LoggerFactory;

import org.springframework.boot.SpringApplication;

import org.springframework.boot.autoconfigure.SpringBootApplication;

import org.springframework.context.ApplicationContext;

import com.cognizant.ormlearn.model.Country;

import com.cognizant.ormlearn.service.CountryService;

@SpringBootApplication

public class OrmLearnApplication {

    private static final Logger LOGGER =

        LoggerFactory.getLogger(OrmLearnApplication.class);

    private static CountryService countryService;

```

```

public static void main(String[] args) {
    ApplicationContext context =
        SpringApplication.run(OrmLearnApplication.class, args);
    countryService = context.getBean(CountryService.class);
    LOGGER.info("Inside main");
    testGetAllCountries();
}

private static void testGetAllCountries() {
    LOGGER.info("Start");
    List<Country> countries = countryService.getAllCountries();
    LOGGER.debug("countries={}", countries);
    LOGGER.info("End");
}
}

```

2) Difference Between JPA, Hibernate and Spring Data JPA

Feature	JPA	Hibernate	Spring Data JPA
Type	Specification	ORM Framework	Spring Framework
Developed By	Jakarta EE	Hibernate Team	Spring
Implementation	No	Yes	Uses JPA
SQL Generation	No	Yes	Yes
Boilerplate Code	More	Less	Very Less
CRUD Support	Manual	Available	Automatic
Repository Support	No	No	Yes (JpaRepository)

JDBC(Without hybernity)

```

Connection con = DriverManager.getConnection(...);

```

```
PreparedStatement ps = con.prepareStatement( "insert into employee values (?, ?)");
ps.setInt(1, 1);

ps.setString(2, "John");
```

Using Hibernate

```
Session session = factory.openSession();

Transaction tx = session.beginTransaction();

session.save(employee);

tx.commit();

session.close();
```

Spring Data JPA

Repository

```
public interface EmployeeRepository
extends JpaRepository<Employee, Integer> {

}
```

Service

```
@Autowired
private EmployeeRepository employeeRepository;

@Transactional
public void addEmployee(Employee employee) {

    employeeRepository.save(employee);

}
```